

AURCHI CHOWDHURY

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EDUCATION

Bangladesh University of Engineering and Technology

Bachelor of Engineering in Computer Science and Engineering | CGPA: 3.88/4.00

Nov 2022 – Present

Dhaka, Bangladesh

Viqarunnisa Noon School and College

High School | GPA: 5.00/5.00 | Board Scholarship

Jan 2009 – Dec 2021

Dhaka, Bangladesh

INTERESTS

Natural Language Processing · Multimodal AI · Computer Vision · Machine Learning · Mechanistic Interpretability

PUBLICATIONS

[Causal Localization of the English Pivot in LLaVA: Mechanistic VLM Analysis and Training-Free Multilingual Steering](#)

Jul 2026

1st Workshop on Multilinguality in the Era of Large Language Models (MeLLM), ACL 2026 | Archival

San Diego, CA

[ACL Anthology](#)

[Where Do Affordances Crystallize? Video vs. Image Self-Supervised Encoders for Embodied Perception](#)

Jun 2026

7th Annual Embodied AI (EAI) Workshop, CVPR 2026 | Non-archival Poster

Denver, CO

(To appear in Workshop website)

WORK EXPERIENCE

Undergraduate Research Intern

On-site

Progya: Adaptive Tutoring Platform | Dept. of CSE, BUET (UGC-funded)

Nov 2025 – May 2026

- Developing an adaptive tutoring platform using Bayesian Knowledge Tracing (BKT), building learning pathways aligned with the Bangladesh Civil Service (BCS) exam syllabus.
- Supervisors: [Md. Saifur Rahman](#), [Md. Shariful Islam Bhuiyan](#)

PROJECTS

[Bengali Long-Form ASR and Diarization](#) | *PyTorch, Whisper, Pyannote, Silero VAD*

Mar 2026

- Tackled low-resource Bengali **Natural Language Processing** by building **WhisperAlign** — a **cross-attention**-guided word-boundary-aware chunker for fine-tuning **Whisper-medium** on Bengali ASR — and domain-adapting **Pyannote**'s segmentation head with **Dual-VAD intersection** for speaker diarization.
- [Code](#) | [Paper](#) | [Slides](#)

[MemoryLane](#) | *PyTorch, ConvNeXt, U-Net, PatchGAN, VGG*

Apr 2026

- Approached grayscale colorization as a **Computer Vision** problem of conditional image generation in CIELab space, building a **ConvNeXt-Base encoder-U-Net decoder** with a **PatchGAN** discriminator; mitigated the gray-mean bias endemic to deep colorization via colorfulness-filtered training and a composite **perceptual + adversarial loss** (VGG + PatchGAN).
- [Code](#) | [Slides](#)

[Scramble](#) | *ATmega32, SG90 servo motors, HC-05 Bluetooth module, Kotlin*

Jul 2025

- Developed a Fully Automated Rubik's Cube Solver capable of generating solutions in a maximum of 20 moves.

[Project Demonstration](#)

[On the Road](#) | *Node, Express, React, PostgreSQL*

Mar 2024

- Built a full-stack travel planning platform covering accommodations, dining, attractions, and tour guides.

ACHIEVEMENTS

- 🏆 11th / 124, DL Sprint 4.0 — Bengali Long-Form ASR & Diarization | BUET CSE Fest 2026 Mar 2026
7th / 124 in ASR and 4th / 124 in Diarization; built **WhisperAlign** for low-resource Bengali ASR and domain-adapted **Pyannote** with **Dual-VAD intersection** for speaker diarization.
- 🌐 1st Runner Up, ResearchSpark 3 Minute Thesis Competition | BUET Apr 2026
Built **BanglaHateCorpus** (168K instances) and **BanHateFormer** — a **Mixture-of-Experts** Transformer achieving 91.33% F1 at 94× smaller scale than BanglaBERT for **NLP-based content moderation**.
- 🌐 1st Runner Up, HULT Prize at BUET Feb 2026
Pitched **ATLAS**, a **multimodal AI** tutoring platform integrating **Vision Transformer OCR**, **Bayesian Knowledge Tracing**, and **BanglaBERT-based NLP grading** for 24M underserved students.
- 🌐 2nd Runners Up, Research Poster Presentation | BUET CSE Fest 2024 Oct 2024
Designed a real-time **multimodal threat detection** system combining **NLP-based rumor classification**, **CNN+RNN** video analysis, and a confidence-scoring fusion module across 16 threat classes.